Roll Number:

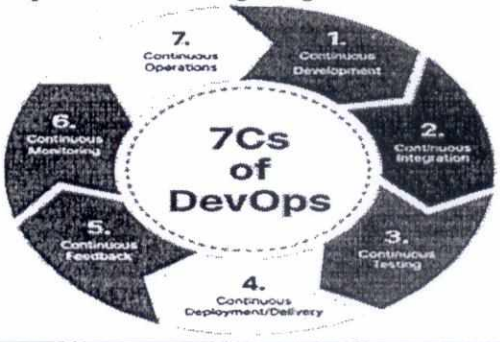
Thapar Institute of Engineering and Technology, Patiala
Computer Science and Engineering Department

BE- CSE/COE (VII Semester) EST — UCS 745: Cloud and DevOps

16 December 2025

Time: 03 Hours; MM: 50 — Faculty Coordinator(s)
Sumit Miglani and Samya Muhuri

Note: Attempt only 4 questions and answer all its parts in the same place

QNO	Questions	Marks	CO	BL
1 a	Implement the Shell Script Program to find out whether the entered number is the Armstrong Number or Not.	5	CO1	L3
1 b	A Group of Students have to develop the Hospital Management System. The Project is large and complex(appointment booking , patient record management, pharmacy, billing and lab report generation). The team chooses to deliver the system module by module so that doctors can use appointment first while other features may be added later i) Identify which SDLC model should be used .Justify it. ii) Discuss the advantages and disadvantages of using the Model. iii) If the teams want to shift their projects to Agile what changes should they make in their process.	6	CO2	L6
1 c	Identify the following icon and explain its role in DevOps 	1.5	CO3	L2
2 a	"*DevOps is not considered to be a Software Development Life Cycle Model*". Discuss the benefits exhibit by DevOps to satisfy the above statement.	6	CO2	L2
2 b	Explain the need of Version Control System. Compare and Contrast the operations of CVCS and DVCS in detail along with the suitable diagram.	6	CO4	L4
2 c	Discuss the usge of 'UAT' in Docker.	0.5	CO6	L2
3 a	How Maven will execute the Project Build .Discuss about the different types of repositories and the dependencies handled by the Maven in detail.	4	CO4	L3
3 b	Interpret the following image 	7	CO1	L3
3 c	Robin is working as a DevOps Engineer in a firm 'Oceania'. He has to prepare the Windows Machine in order to carry out the project build. What steps he should follow to install the apache-maven-3.9.0-bin.zip?	1.5	CO3	L6

4 a	Consider Debian based Linux distribution having Java installed and file ABC.txt present and created by user Alex. Implement the tasks i) Uninstall the Java from the Machine ii) Display first 100 lines of file as an output iii) Change the owner and group of file to Bob at same time iv) Display default permissions associated with file in proper syntax v) Delete the complete profile of user Alex from the System vi) Which Package Manager is used.	5.5	CO2	L5
4 b	List any five features of DevOps.	5	CO1	L1
4 c	Perform the comparative analysis of Elasticity and Scalability in the domain of the Cloud Computing	2	CO3	L4
5 a	Describe about POM.xml and various types of Goals in MAVEN	5	CO4	L2
5 b	Discuss the benefits of the cloud computing over the infrastructure computing in detail	6	CO2	L2
5 c	Consider the company spring framework.org working on the project spring-core app with current version number 5.3.23.Use the information to write the syntax to add it as a dependency in POM.xml .	1.5	CO3	L3
6 a	David have to prepare the Docker Machine in Amazon Linux What steps he must perform to install the Docker. The machine is running the image name as 'Forever-Always'. Write the syntax to run the Docker container in the detached mode and to remove the Docker image	5	CO6	L6
6 b	Explain the operations involved between the Working Tree and the Remote Repository URL : http://Demo_First in forwarding the file XYZ.txt.	6	CO5	L2
6 c	Distinguish the Cloud Service Models diagrammatically	1.5	CO3	L4
7 a	Design a scenario to show case the conflict resolution in GITHUB.	5	CO5	L6
7 b	Formulate the communication Steps between the Docker client and Docker daemon in retrieving and running the 'hello-world' image present in the Docker Hub	5.5	CO6	L6
7 c	Discuss the significance of CMD and COPY keywords in DOCKER file .	2	CO5	L2